6.Naïve Bayes algorithm

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score, classification_report

iris = load_iris()
X = iris.data
y = iris.target


X_train, X_test, y_train, y_test = train_test_split(
   X, y, test_size=0.3, random_state=42
)

model = GaussianNB()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Actual Classes:")
print(y_test)

print("\nPredicted Classes:")
print(y_pred)

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nAccuracy:")
print(accuracy_score(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

OUTPUT:

```
Actual Classes:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Predicted Classes:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 2 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Confusion Matrix:
[[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]

Accuracy:
0.9777777777777777

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        19
           1       1.00      0.92      0.96        13
           2       0.93      1.00      0.96        13

    accuracy                           0.98        45
   macro avg       0.98      0.97      0.97        45
weighted avg       0.98      0.98      0.98        45
```